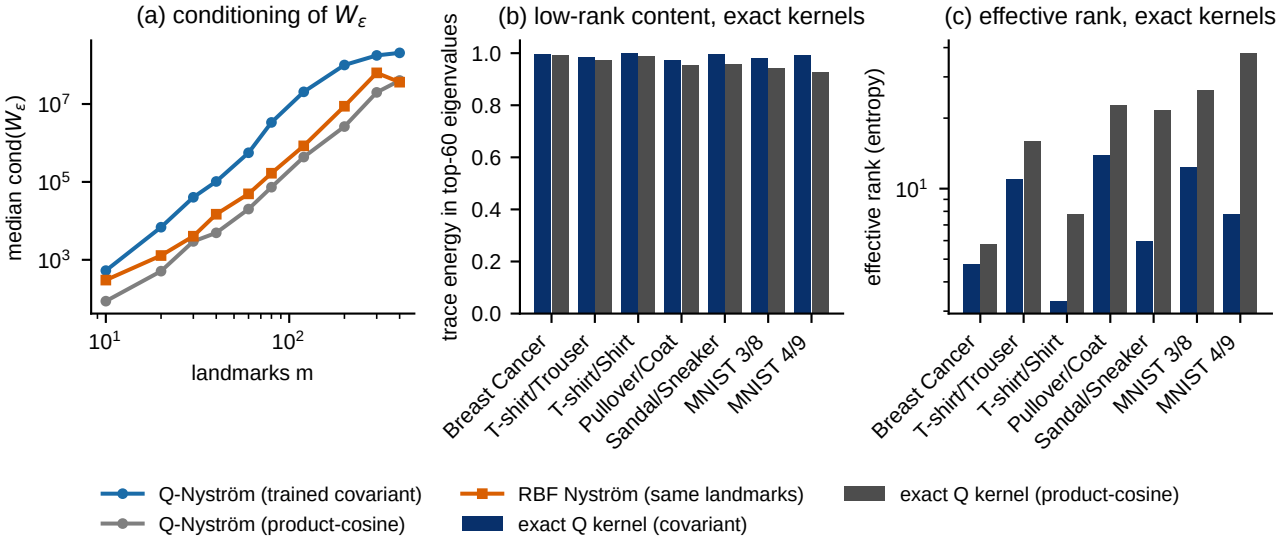
(a) conditioning of W_ε
median cond(W_ε)
10^7
10^5
10^3
10^1
10^2
landmarks m
(b) low-rank content, exact kernels
trace energy in top-60 eigenvalues
1.0
0.8
0.6
0.4
0.2
0.0
Breast Cancer
T-shirt/Trouser
T-shirt/Shirt
Pullover/Coat
Sandal/Sneaker
MNIST 3/8
MNIST 4/9
(c) effective rank, exact kernels
effective rank (entropy)
10^1
Breast Cancer
T-shirt/Trouser
T-shirt/Shirt
Pullover/Coat
Sandal/Sneaker
MNIST 3/8
MNIST 4/9
Q-Nyström (trained covariant)
Q-Nyström (product-cosine)
RBF Nyström (same landmarks)
exact Q kernel (covariant)
exact Q kernel (product-cosine)